

OPTT Analysis

```
#pull OPTH ticker data for all time since going public.
#date format: YYYY-MM-DD

optt <- tq_get("OPTT",
               get = "stock.prices",
               from = "2007-04-25", #went public on NASDAQ April 25th 2007
               to = "2026-06-21") #at time of download, update as needed
```

```
#preview data
```

```
head(optt)
```

```
# A tibble: 6 x 8
```

	symbol	date	open	high	low	close	volume	adjusted
	<chr>	<date>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	OPTT	2007-04-25	4000	4000	3548	3568	15321	3568
2	OPTT	2007-04-26	3510	3590	3210	3336	6585	3336
3	OPTT	2007-04-27	3348	3436	3110	3110	1157	3110
4	OPTT	2007-04-30	3150	3196	2850	2868	1568	2868
5	OPTT	2007-05-01	2968	2968	2760	2860	1623	2860
6	OPTT	2007-05-02	2874	3250	2874	3184	1445	3184

```
#summary table
```

```
summary(optt)
```

symbol	date	open	high
Length:4819	Min. :2007-04-25	Min. : 0.12	Min. : 0.13
Class :character	1st Qu.:2012-02-02	1st Qu.: 1.04	1st Qu.: 1.09
Mode :character	Median :2016-11-15	Median : 44.60	Median : 47.00

	Mean	:2016-11-17	Mean	: 461.69	Mean	: 475.95	
	3rd Qu.:	2021-08-30	3rd Qu.:	644.00	3rd Qu.:	672.00	
	Max.	:2026-06-18	Max.	:4000.00	Max.	:4000.00	
low	close		volume		adjusted		
Min. :	0.12	Min. :	0.130	Min. :	0	Min. :	0.130
1st Qu.:	1.00	1st Qu.:	1.045	1st Qu.:	285	1st Qu.:	1.045
Median :	42.00	Median :	44.000	Median :	7710	Median :	44.000
Mean :	445.57	Mean :	459.070	Mean :	1974392	Mean :	459.070
3rd Qu.:	619.00	3rd Qu.:	639.000	3rd Qu.:	350200	3rd Qu.:	639.000
Max. :	3550.00	Max. :	3652.000	Max. :	616465100	Max. :	3652.000

```
#tidy datatable

#rename variables to be more
optt_tidy <- optt |>
  rename(open_price = open,
         high_price = high,
         low_price = low,
         close_price = close,
         Volume = volume,
         Adjusted = adjusted) |>

#break date into day, month, year columns
  separate(date, into = c("Year", "Month", "Day"), sep = "-", convert = TRUE) |>

#add month name column
  mutate(month_name = month.name[Month]) |>

#place month name column after month number
  relocate(month_name, .after = Month) |>

#create new variables
  mutate(
    #intra-day volatility
    idv = (high_price-low_price),
    #previous close price
    prev_close = lag(close_price),
    #absolute maximum upward movement
    max_up = (abs(high_price - prev_close)),
    #absolute maximum downward movement
    max_down = (abs(low_price - prev_close))
  )
```

#idv, max_up, max_down will be used to calculate True Range, which will be used to calculate